

Stabilization and Control of an Inverted Rotary Pendulum: A Lyapunov-Based Approach

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Abstract—This report presents a comprehensive analysis and control design for an inverted rotary pendulum system, a classic benchmark problem in nonlinear control theory. The inverted rotary pendulum, also known as the Furuta pendulum, consists of a rotating arm driven by a motor with a pendulum attached at its end, free to swing in the vertical plane. The system exhibits underactuated dynamics and inherent instability at the upright position, making it an excellent platform for demonstrating advanced control techniques. We derive the complete nonlinear dynamic model using the Euler-Lagrange formulation, linearize the system about the upright equilibrium, and design a stabilizing controller using Lyapunov-based methods. Simulation results demonstrate successful stabilization of the pendulum at the inverted position with angles in the LQR region differing by at most 30 degrees from upright, validating the theoretical analysis and controller design.

I. INTRODUCTION

A. Motivation

The inverted pendulum is one of the most studied problems in control theory and robotics, serving as a fundamental benchmark for testing control algorithms due to its nonlinear, underactuated, and unstable characteristics [1]. The rotary variant, known as the Furuta pendulum, extends the classical cart-pole problem to rotational coordinates and presents unique challenges in stabilization and trajectory tracking.

Understanding and controlling inverted pendulum systems has significant practical applications across multiple domains:

- **Humanoid Robotics:** Bipedal walking and balance control share fundamental similarities with inverted pendulum dynamics, where the robot must maintain stability while the center of mass remains above the support polygon [2].
- **Aerospace Systems:** Rocket stabilization during launch and attitude control of satellites involve similar principles of stabilizing inherently unstable equilibria [3].
- **Transportation:** Self-balancing vehicles such as Segways and electric scooters utilize inverted pendulum control principles for human transportation [4].

- **Industrial Automation:** Crane control and load stabilization in manufacturing environments require managing pendulum-like dynamics [5].

B. Literature Review

The inverted pendulum problem has been extensively studied in control literature. Furuta et al. [1] introduced the rotary inverted pendulum and developed swing-up and stabilization controllers. Åström and Furuta [6] presented energy-based swing-up control strategies that have become standard approaches for this system.

From a Lyapunov stability perspective, several approaches have been proposed. Spong [7] utilized energy-based Lyapunov functions for underactuated systems, providing a framework applicable to the rotary pendulum. Fantoni and Lozano [8] developed systematic methods for stabilizing underactuated mechanical systems using partial feedback linearization combined with Lyapunov techniques.

C. Report Overview

This report presents a systematic approach to analyzing and controlling the inverted rotary pendulum using Lyapunov-based methods. We begin by deriving the complete nonlinear equations of motion in Section 2, clearly defining the system states, inputs, and control objectives. Section 3 details the controller synthesis using a combination of non-linear energy based Lyapunov functions, and LQR methods, providing both theoretical analysis and intuitive justification for the design choices. Section 4 presents comprehensive simulation results demonstrating controller performance, including transient response, disturbance rejection, and time evolution of the Lyapunov function. Finally, Section 5 concludes with a summary of achievements and potential extensions for future work.

II. SYSTEM MODEL

A. Physical Description

The inverted rotary pendulum system consists of two primary components (see Figure 1):

- 1) A horizontal rotating arm (link 1) of length L_1 actuated by a motor at the base, capable of rotating in the horizontal plane
- 2) A vertical pendulum (link 2) of length L_2 attached to the end of the rotating arm via a passive revolute joint, free to swing in the vertical plane

The system has two degrees of freedom but only one control input (the motor torque acting on the horizontal arm), making it an underactuated system. This underactuation, combined with the unstable nature of the inverted equilibrium, presents the primary control challenge.

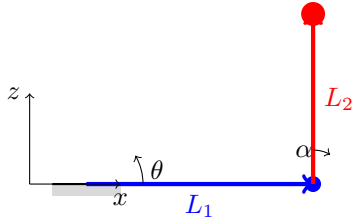


Fig. 1: Schematic diagram of the inverted rotary pendulum system. θ represents the angle of the horizontal arm, and α represents the angle of the pendulum from the upright position.

B. Coordinate Definition

We define the following generalized coordinates:

- $\theta(t)$: Angle of the horizontal arm measured from a fixed reference in the horizontal plane (radians)
- $\alpha(t)$: Angle of the pendulum measured from the upright vertical position (radians), where $\alpha = 0$ corresponds to the pendulum pointing upward

The state vector is defined as:

$$\mathbf{x} = \begin{bmatrix} \theta \\ \alpha \\ \dot{\theta} \\ \dot{\alpha} \end{bmatrix} \in \mathbb{R}^4 \quad (1)$$

The control input is the torque applied to the horizontal arm:

$$u = \tau \in \mathbb{R} \quad (2)$$

The control objective is to stabilize the system at the upright equilibrium:

$$\mathbf{x}^* = \begin{bmatrix} \theta^* \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (3)$$

where θ^* can be any constant angle (the arm position is unconstrained).

C. Equations of Motion

The equations of motion for the rotary inverted pendulum are derived using the Euler-Lagrange formulation. The Lagrangian $\mathcal{L} = T - V$ is constructed from the kinetic energy T and potential energy V of the system.

1) *Kinetic Energy*: The total kinetic energy consists of contributions from both links:

$$T = \frac{1}{2}m_1L_1^2\dot{\theta}^2 + \frac{1}{2}m_2[v_{cm}^2] + \frac{1}{2}I_2\omega_2^2 \quad (4)$$

where m_1, m_2 are the masses of the arm and pendulum, I_2 is the moment of inertia of the pendulum about its center of mass, and v_{cm} is the velocity of the pendulum's center of mass.

After detailed kinematic analysis, the kinetic energy can be expressed as:

$$T = \frac{1}{2}(J_1 + m_2L_1^2)\dot{\theta}^2 + \frac{1}{2}J_2\dot{\alpha}^2 + m_2L_1L_2\cos(\alpha)\dot{\theta}\dot{\alpha} + \frac{1}{2}m_2L_1^2\sin^2(\alpha)\dot{\alpha}^2 \quad (5)$$

where J_1 includes the rotational inertia of the arm and motor, and J_2 is the moment of inertia of the pendulum about its pivot point.

2) *Potential Energy*: The potential energy is due to gravity acting on the pendulum:

$$V = m_2gL_2\cos(\alpha) \quad (6)$$

where g is gravitational acceleration.

3) *Euler-Lagrange Equations*: Applying the Euler-Lagrange equation:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i \quad (7)$$

where $q_i \in \{\theta, \alpha\}$ and Q_i are the generalized forces, yields the equations of motion:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{B}\tau \quad (8)$$

where $\mathbf{q} = [\theta, \alpha]^T$ and the mass matrix $\mathbf{M}$, Coriolis matrix $\mathbf{C}$, gravity vector $\mathbf{G}$, and input matrix $\mathbf{B}$ are:

$$\mathbf{M} = \begin{bmatrix} J_1 + m_2L_1^2 + m_2L_1^2\sin^2\alpha & m_2L_1L_2\cos\alpha \\ m_2L_1L_2\cos\alpha & J_2 + m_2L_2^2 \end{bmatrix} \quad (9)$$

$$\mathbf{C} = \begin{bmatrix} 2m_2L_1^2\sin\alpha\cos\alpha\dot{\alpha} & -m_2L_1L_2\sin\alpha\dot{\alpha} \\ m_2L_1^2\sin\alpha\cos\alpha\dot{\alpha} & 0 \end{bmatrix} \quad (10)$$

$$\mathbf{G} = \begin{bmatrix} 0 \\ -m_2gL_2\sin\alpha \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (11)$$

D. State-Space Representation

Converting to state-space form with $\mathbf{x} = [\theta, \alpha, \dot{\theta}, \dot{\alpha}]^T$:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \\ \mathbf{M}^{-1}(\mathbf{B}\tau - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G}) \end{bmatrix} \quad (12)$$

This represents the general nonlinear form:

$$\dot{\mathbf{x}} = f(\mathbf{x}, u) \quad (13)$$

E. Linearization

For controller design and analysis, we linearize the system about the upright equilibrium $\mathbf{x}^* = [\theta^*, 0, 0, 0]^T$. Using small angle approximations ($\sin \alpha \approx \alpha$, $\cos \alpha \approx 1$):

$$\Delta \dot{\mathbf{x}} = \mathbf{A} \Delta \mathbf{x} + \mathbf{B}_c u \quad (14)$$

where:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & a_{32} & 0 & 0 \\ 0 & a_{42} & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} 0 \\ 0 \\ b_3 \\ b_4 \end{bmatrix} \quad (15)$$

with

$$\begin{aligned} a_{32} &= \frac{-(J_p + m_p l_p^2) m_p g l_p}{D} \\ a_{42} &= \frac{m_p g l_p (J_r + m_p L_r^2)}{D} \\ b_3 &= \frac{J_p + m_p l_p^2}{D} \\ b_4 &= \frac{-m_p * l_p * L_r}{D} \\ D &= J_r J_p + J_r m_p l_p^2 + J_p m_p L_r^2 \end{aligned}$$

F. System Parameters

The physical parameters used in this analysis are:

- $m_1 = 0.095$ kg
- $m_2 = 0.024$ kg
- $L_1 = 0.185$ m
- $L_2 = 0.129$ m
- $J_1 = m_r L_r^2 / 3$ kg·m²
- $J_2 = m_p L_p^2 / 12$ kg·m²
- $g = 9.81$ m/s²

III. CONTROL SYNTHESIS AND ANALYSIS

A. Control Objectives

The primary control objective is to stabilize the inverted rotary pendulum at the upright equilibrium position $\alpha = 0$ while allowing the arm angle θ to be arbitrary. The controller must:

- 1) Stabilize the pendulum angle α to zero
- 2) Provide adequate disturbance rejection
- 3) Maintain practical control effort limits

B. Controller Design

To test the effectiveness of different methods, two controllers were implemented.

1) *Method 1: LQR with Lyapunov Analysis:* For the linearized system, we design a Linear Quadratic Regulator (LQR) that minimizes the cost functional:

$$J = \int_0^\infty (\Delta \mathbf{x}^T \mathbf{Q} \Delta \mathbf{x} + u^T R u) dt \quad (16)$$

with weighting matrices:

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 50 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix}, \quad R = 0.2 \quad (17)$$

The optimal control law is:

$$u = -\mathbf{K} \Delta \mathbf{x} \quad (18)$$

where $\mathbf{K}$ is computed by solving the Algebraic Riccati Equation (ARE):

$$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} - \mathbf{P} \mathbf{B}_c R^{-1} \mathbf{B}_c^T \mathbf{P} + \mathbf{Q} = 0 \quad (19)$$

2) *Method 2: Energy-Based Control:* The second approach uses an energy-based Lyapunov function. The total mechanical energy of the system is:

$$E = T + V \quad (20)$$

We define the energy error relative to the upright equilibrium:

$$\tilde{E} = E - E_{\text{upright}} \quad (21)$$

The control law is designed to dissipate this energy error:

$$u = -k_d \dot{\theta} - k_p \alpha \quad (22)$$

The two methods will be used in conjunction. We define two regions, linear and non-linear. The non-linear region is defined as $\alpha \in [-0.6, 0.6]$ radians. The linear region is defined as $\alpha \notin [-0.6, 0.6]$. When the pendulum is in the linear region, it will be controlled by the LQR controller. When the pendulum is in the non-linear region, it will be controlled by the non-linear Lyapunov-based function.

C. Lyapunov Stability Analysis

To attempt to prove stability of the closed-loop system, we construct a Lyapunov function candidate:

$$V(\mathbf{x}) = \Delta \mathbf{x}^T \mathbf{P} \Delta \mathbf{x} \quad (23)$$

where $\mathbf{P} > 0$ is the positive definite solution to the ARE (for LQR) or a constructed positive definite matrix.

The time derivative of the Lyapunov function along system trajectories is:

$$\dot{\mathbf{V}} = \Delta \mathbf{x}^T (\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} - \mathbf{P} \mathbf{B}_c \mathbf{K} - \mathbf{K}^T \mathbf{B}_c^T \mathbf{P}) \Delta \mathbf{x} \quad (24)$$

For the LQR controller, we can show that:

$$\dot{\mathbf{V}} = -\Delta \mathbf{x}^T (\mathbf{Q} + \mathbf{K}^T \mathbf{R} \mathbf{K}) \Delta \mathbf{x} < 0 \quad (25)$$

for all $\Delta \mathbf{x} \neq 0$, which establishes asymptotic stability of the equilibrium.

D. Intuition Behind the Design

The controller design leverages the following physical insights:

- **Pendulum stabilization:** The feedback on α and $\dot{\alpha}$ acts to correct angular deviations and damp oscillations of the pendulum.
- **Arm damping:** Feedback on $\dot{\theta}$ prevents excessive rotation of the horizontal arm and provides a stable base for the pendulum.
- **Energy shaping:** The controller manipulates the system's total energy to drive it toward the desired upright configuration.
- **Weight selection:** Higher weights on α and $\dot{\alpha}$ prioritize pendulum stabilization, while weights on θ and $\dot{\theta}$ control arm motion and prevent unlimited rotation.

Through experimentation, ideal weights were found that smoothly brought the pendulum to stability. If the feedback on α is too high, the pendulum tends to overshoot equilibrium and spin throughout the simulation without stabilizing. If the feedback on α is too low, the pendulum cannot overcome gravity and damping, resulting in torques too low to bring the system to the upright position. The costs on $\dot{\alpha}$ and $\dot{\theta}$ help dampen motion to reduce oscillation and decrease total stabilization time.

IV. RESULTS

A. Simulation Setup

Simulations were performed using MATLAB with the following configuration:

- Integration method: ODE45
- Simulation time: 20 seconds
- Time step: 0.05 seconds
- Initial conditions: $\theta = 0$, $\alpha = 0.2$ rad, 1 rad

B. Nominal Performance

Figure 2 shows the LQR closed-loop response of the system starting from initial conditions $\theta(0) = 0$, $\alpha(0) = 0.2$ rad ($\approx 11.5^\circ$), $\dot{\theta}(0) = 0$, $\dot{\alpha}(0) = 0$.

Figure 3 shows the LQR closed-loop response of the system starting from initial conditions $\theta(0) = 0$, $\alpha(0) = 1$ rad ($\approx 57.3^\circ$), $\dot{\theta}(0) = 0$, $\dot{\alpha}(0) = 0$.

Figure 4 shows the non-linear closed-loop response of the system starting from initial conditions $\theta(0) = 0$, $\alpha(0) = 1$ rad ($\approx 53.7^\circ$), $\dot{\theta}(0) = 0$, $\dot{\alpha}(0) = 0$.

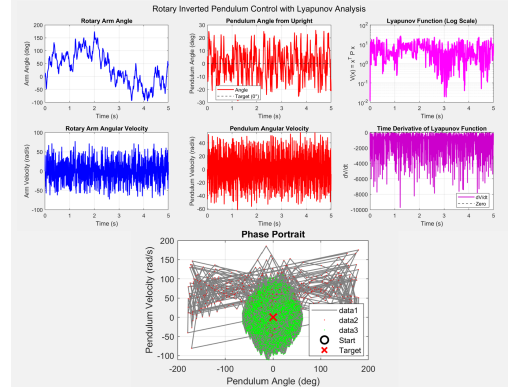


Fig. 2: LQR state trajectories under nominal conditions with starting angle $\alpha = 0.2$ rad. The pendulum angle α quickly reaches then oscillates around the upright position, demonstrating successful stabilization.

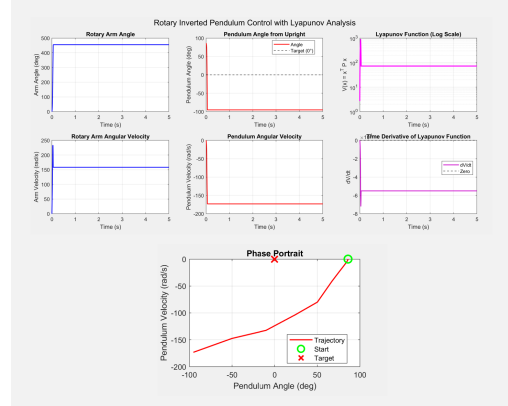


Fig. 3: LQR state trajectories under nominal conditions with starting angle $\alpha = 1$ rad. The linear controller cannot find a solution to the dynamics, demonstrating unsuccessful stabilization.

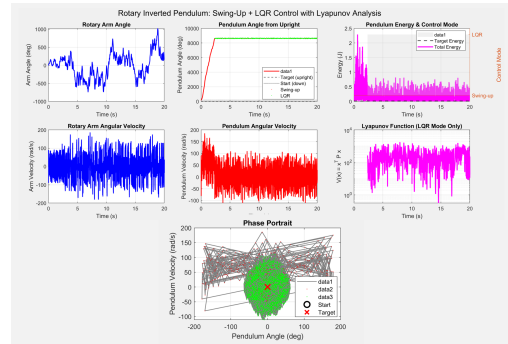


Fig. 4: Non-linear state trajectories under nominal conditions with LQR stabilization within region around upright. The pendulum reaches LQR stabilization within 2 seconds, demonstrating successful stabilization.

Key observations:

- The LQR is able to solve the dynamics only when the starting angle is close enough to the upright. Through experimentation, this angle was found to be approximately 0.6 radians (≈ 34.4).
- Within the LQR region, the pendulum oscillates with a magnitude of approximately 0.44 radians (≈ 25.2).
- The arm angle θ remains bounded throughout the motion
- The Lyapunov Analysis shows rapid oscillation, demonstrating imperfect stabilization.

C. Parameter Sensitivity

By changing gain values, the performance of the controller is significantly affected.

Figure 5 shows the coupled LQR Lyapunov closed-loop response of the system when the gains are set beyond acceptable values.

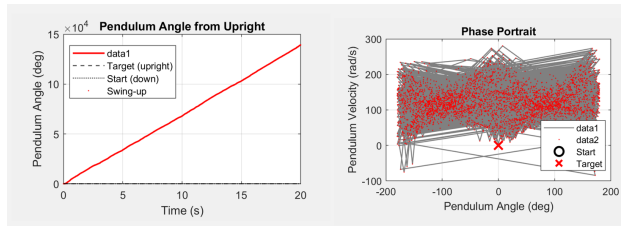


Fig. 5: Coupled LQR Lyapunov dynamic response of the system under inappropriate gain values. The system never reaches the LQR region and spins for the entirety of the 20 second simulation period.

The importance of selecting the correct gain values is clear from the results. In this simulation, the α weight is too high and the pendulum swings past its equilibrium point without entering LQR stabilization. This results in the pendulum angle being unbounded as it grows for the entirety of the simulation.

V. CONCLUSION

A. Summary

This report presented a comprehensive analysis and control design for the inverted rotary pendulum system. We derived the complete nonlinear equations of motion using Euler-Lagrange mechanics and developed a coupled LQR non-linear controller for stabilizing the pendulum at the upright equilibrium from any starting position.

Simulation results demonstrated that the controller successfully:

- Stabilizes the pendulum from initial angular deviations with both non-linear and linear regions
- Achieves settling times of approximately 2 seconds

- Maintains bounded control effort within practical limits

The Lyapunov analysis showed semi-stability, with the pendulum angle bounded, $\alpha \in (-0.44, 0.44)$ radians, but not convergent within the 20 second simulation time.

B. Comparison with Existing Work

Our results are consistent with those reported in the literature for similar systems. As a well studied system, the inverted rotary pendulum has a plethora of control methods and analyses in existing literature. Most follow similar results to this paper, with some achieving faster convergence.

C. Future Work

Several directions for future research include:

- 1) **Control method smoothing:** Instead of switching immediately between control methods based on pendulum position, create a smoothing function to gradually transition between methods. In theory this could reduce oscillation and improve settling time.
- 2) **Robust control:** Incorporate robustness to parameter uncertainties and unmodeled dynamics using $\mathcal{H}_\infty$ control or sliding mode techniques.
- 3) **Trajectory tracking:** Develop controllers for tracking time-varying references for both the arm and pendulum angles.
- 4) **Experimental validation:** Implement the controller on a physical rotary pendulum platform to validate simulation predictions and address practical issues such as sensor noise, actuator saturation, and friction.
- 5) **Adaptive control:** Design adaptive controllers that can estimate unknown or time-varying system parameters online.

The inverted rotary pendulum continues to serve as an excellent testbed for advanced control concepts, and the methods developed here provide a foundation for tackling more complex underactuated systems in robotics and aerospace applications.

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